

Adam MOUNIR

AI Research Student

+33 6 51 30 63 33 | adam.mounir@efrei.net | [LinkedIn](#) | [Github](#) | [Portfolio](#)

EDUCATION

Centrale Supélec - Université Paris-Saclay

MSc. - Artificial Intelligence

- **Courses S1** : Optimization for ML - Decision Modelling - Machine Learning - Deep Learning

MVA - ENS Paris-Saclay

- **Courses S2** : Graph ML - Stochastic Optimization - Advanced Deep Learning - NLP - Reinforcement Learning

EFREI - Paris Panthéon-Assas

MEng. - Diplôme d'Ingénieur in Data & Artificial Intelligence

- **Courses** : Convex Optimization - Stats/Probas - Graph Theory - Machine / Deep Learning - Reinforcement Learning - NLP

University of California, Irvine

Exchange Student, MEng. in Computer Science

- **Courses** : Mathematics for Cryptography - Java Programming - Computer Architecture - Network Protocols

PROFESSIONAL EXPERIENCE

Inria - TAU (Université Paris-Saclay)

AI Research - Intern

Researching deep-learning architectures for EEG decoding in [Braindecode](#) and Growing Neural Networks during training in [EEGrow](#)

- Design transformer, graph-neural-network, convolutional and Riemannian models for neural decoding using open EEG datasets
- Co-organize the NeurIPS 2026 [EEG decoding competition](#) (Meta, top academic labs) building its benchmark infrastructure
- Porting in-training architecture growth (gromo) to EEG decoding: conv-junction width auto-sized during training rather than tuned

Thales

AI Research - Intern

Researched global-scale Root Cause Analysis by distilling multi-step reasoning capabilities from LLMs into specialized architectures

- Designed a supervised alignment pipeline to map expert causal trajectories from logs into internalized reasoning schemas
- Distilled causal traces via GPT-4o to fine-tune a SFT pipeline using Llama3.1 (QLoRA) targeting latent reasoning convergence
- Achieved a 0.64 F1-score in RCA prediction and established a validation protocol for Out-of-Distribution scenarios

Société Générale

AI Engineer - Apprentice

Detected semantic discrepancies between unstructured documents disclosures and structured multi-provider data

- Automated cross-source data alignment and causal influence tracing within hierarchical scoring models
- Engineered a Zero-shot NER pipeline using Claude 3.5 Sonnet (RAG) and a Directed Graphs for topological impact analysis

RESEARCH PROJECTS

Braindecode

Open-Source Contribution | [Github Link](#)

Maintaining and extending the reference PyTorch/skorch library for end-to-end neural time-series decoding

- Implementing transformer, graph-neural-network and Riemannian architectures for EEG/MEG signal decoding
- Building reproducible benchmarking pipelines over open neurophysiology datasets (MOABB integration)
- Contributing tested and documented modules to a community-driven library used across academic and clinical labs

Nullflow : Neuro-AI for Complementary Learning Systems

Personal Research Project | [Preprint Link](#)

Tackling catastrophic forgetting in Deep Neural Networks during non-stationary sequential data streams

- Designing bio-inspired architectures optimizing the stability-plasticity trade-off via CLS Theory & Null-Space Projection
- Built a Wake-Sleep/Flow Matching framework in PyTorch for NREM/REM latent generative replay (~2.1M parameters)
- Outperformed 8 CL baselines by +18.8pp via 6-step ODE latent replay;

ACHIEVEMENTS

Google Deepmind x {Tech : Europe}

AI Hackathon - Big Berlin Hack

Selected (Researcher track) among 300+ builders to develop a neuroscience-oriented AI application using frontier multimodal models, competing for €50k+ in prizes

Microsoft x Finelog-Biseum

AI Hackathon - Azure AI Platforms

Built a context-aware HR chatbot on Azure (FastAPI, CosmosDB, Azure OpenAI) with RAG over French labor law documents and role-conditioned response generation via BFS traversal of organizational hierarchies

PERSONAL INTERESTS & SKILLS

Hobbies : Running, Swimming, Reading, Piano

Languages : French (native), English (C1), Spanish (B2)

Programming Languages : Python, C/Assembly 68000, Java/C#/Scala, MATLAB, HTML/CSS/JavaScript, Oracle, SQL